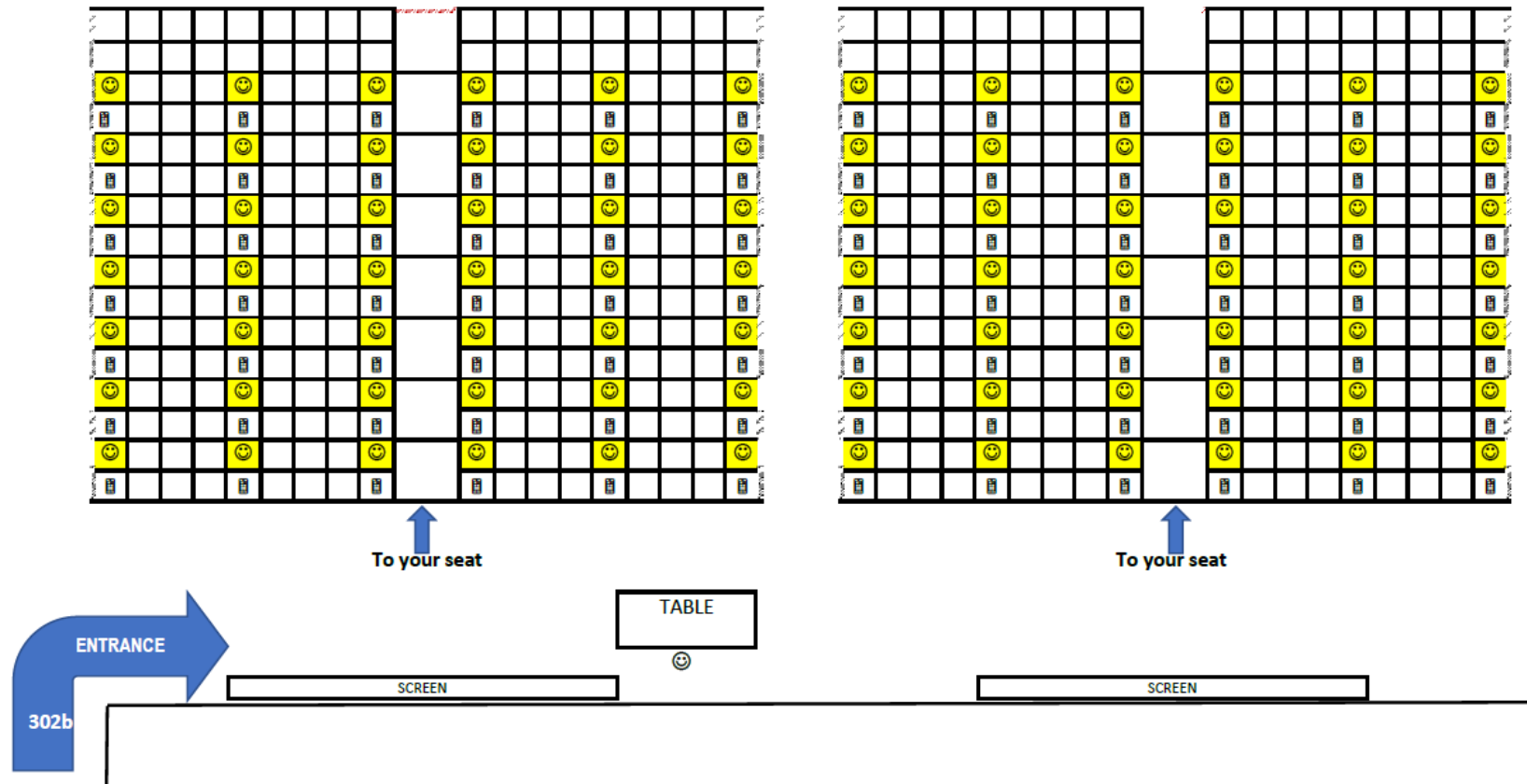


Be prepared for the Mid-semester test!

- Read the **test procedure document** thoroughly
- Check if you are not excluded from the **participant list** and find out the **start time of your session**
- Check the **seating plan** in the auditorium
- Repeat the **topics of all previous lectures**, review the slides and AoA chapters
- **Compile, run and modify all the code examples** provided with the lecture materials



Come on time for your session – you will find the start time on the participant list!

You only need:

- student **ID card**
- a good **pen** (black or blue)

Your **phone and student ID** must be placed **in the previous row on the table** in front of your seat

You must **remain in your seat for the entire 15 minutes**, even if the answers have been completed earlier

CAPITAL BLOCK LETTERS!

DPI343 Computer Organization and Assembly Language

Midterm Test 1

Surname, name SMITH JOHN St. ID 231NDB123 Date 28/10/2025
PLEASE WRITE HERE IN CAPITAL BLOCK LETTERS ONLY!

Open questions only when permission is given

Write your answer exactly as requested in the question

Important topics

- Hexadecimal representation of integer numeric values (signed and unsigned) in registers and memory
- Execution of simple instructions including simple arithmetic calculations
- Loop instruction, nested loops
- Array and text processing
- Procedures, CALL and RET instructions, parameter passing

Example 1

What internal code of the register AX (hexadecimal) will be after execution of the following instructions?

```
.data
vect    word    -1, 2, -3, 4, 5, -6, 7, 8 ; array given

.code
start   proc
        mov ecx, 4
        mov esi, 0
        mov ax, 0
m:      cmp vect[esi], 0
        jge next
        add ax, vect[esi]
next:   add esi, 2
        loop    m
; ::::::::::
start endp
        end start
```

ANSWER (hexadecimal digits without any other symbols):

FFFC

Example 2

What internal code of the memory field **Res** will be after execution of the following instructions?

```
.data
val    byte -1, 2, -3, 4, 5, -6, 7, 8 ; array given
Res    word 0

.code
start  proc
        mov ecx, 4
        mov esi, 0
        mov ax, 0
m:      add al, val[esi]
        add esi, 2
        loop m
        mov Res, ax
; ::::::::::
start endp
        end start
```

ANSWER (hexadecimal digits without any other symbols):

Res	Res+1
08	00

Example 3

Integer numeric value N in word format is defined.

N word 266

Write internal hexadecimal code of each of the bytes starting from the address N.

ANSWER (hexadecimal value):

N	N+1
0A	01

Example 4

General purpose register EAX contains internal code (hexadecimal):

000000FB

What the decimal value of the register AX will be after execution of the following instructions?

```
add    eax, 4
mov    ah, 0FFh
add    ax, 15
```

ANSWER (decimal value):

14

Example 5

What are the values of the EBX and EDX registers after execution of the following instructions?

```
mov    edx,256
push   edx
mov    eax,-10
push   eax
add    eax,edx
push   eax
pop    edx
pop    ebx
add    ebx,edx
```

ANSWER (decimal values): EBX = **236** EDX = **246**

Example 7

Fragment of the code to find number of the asterisk symbols in the text string is given. Write missing instruction(s) in the box to complete this code.

```
.data
string  byte      '012*456*891*ABC', 0    ; text string
```

```
.code
start  proc
        mov eax, 0      ; counter = 0
        mov esi, 0      ; index of the symbol to check
check:  cmp string[esi], 0
        je  done
        cmp string[esi], '*'
```

```
        jne next
        inc eax
```

```
next:   inc esi
        jmp check
```

```
done:   ; number of asterisk symbols is in eax
```

Example 8

Procedure Zeros receives one doubleword parameter via stack and returns the number of bits which contain zero in this doubleword in the register EAX.

```
Zeros proc near stdcall
    push ebp
    mov  ebp, esp
    push ebx
    push ecx

    ; MISSING INSTRUCTION

    xor  ebx, ebx
    mov  ecx, 32
t:    test eax, 1
      jnz  next
      inc  ebx
next: shr  eax, 1
      loop t
    mov  eax, ebx
    pop  ecx
    pop  ebx
    pop  ebp
    ret  4
Zeros endp
```

Which one is a correct instruction to insert in the line of code marked by the comment?

- A. mov eax, [ebp+12]
- B. mov eax, [esp+8]
- ☒ C. mov eax, [ebp+8]
- D. mov ecx, [ebp+8]
- E. None of the above

Example 9

Procedure Zeros receives one doubleword parameter via stack and returns in the register EAX number of bits which contain zero in this doubleword.

```
Zeros proc near stdcall
```

```
    push ebp
    mov  ebp, esp
    push ebx
    push ecx
    mov  eax, [ebp+8]
    xor  ebx, ebx
    mov  ecx, 32
t:    test eax, 1
      jnz next
      inc  ebx
next: shr  eax, 1
      loop t
    mov  eax, ebx
    pop  ecx
    pop  ebx
    pop  ebp
    ret  4
```

```
Zeros endp
```

Write instructions to call this procedure and count zero bits in the doubleword Beta defined as

```
Beta    dword    15
```

ANSWER:

```
push    offset Beta
call     Zeros
```

Example 10

Array of five integer numeric values (in word format) is defined. Each of the five values is stored in two bytes. Address of the first element of the array is X.

Write the contents of the memory byte with address X+5.

X word 1, -1, -2, 42, 0

Write content of the memory byte as two hexadecimal digits (two symbols without any other symbols).

ANSWER: FF

Explanation:

